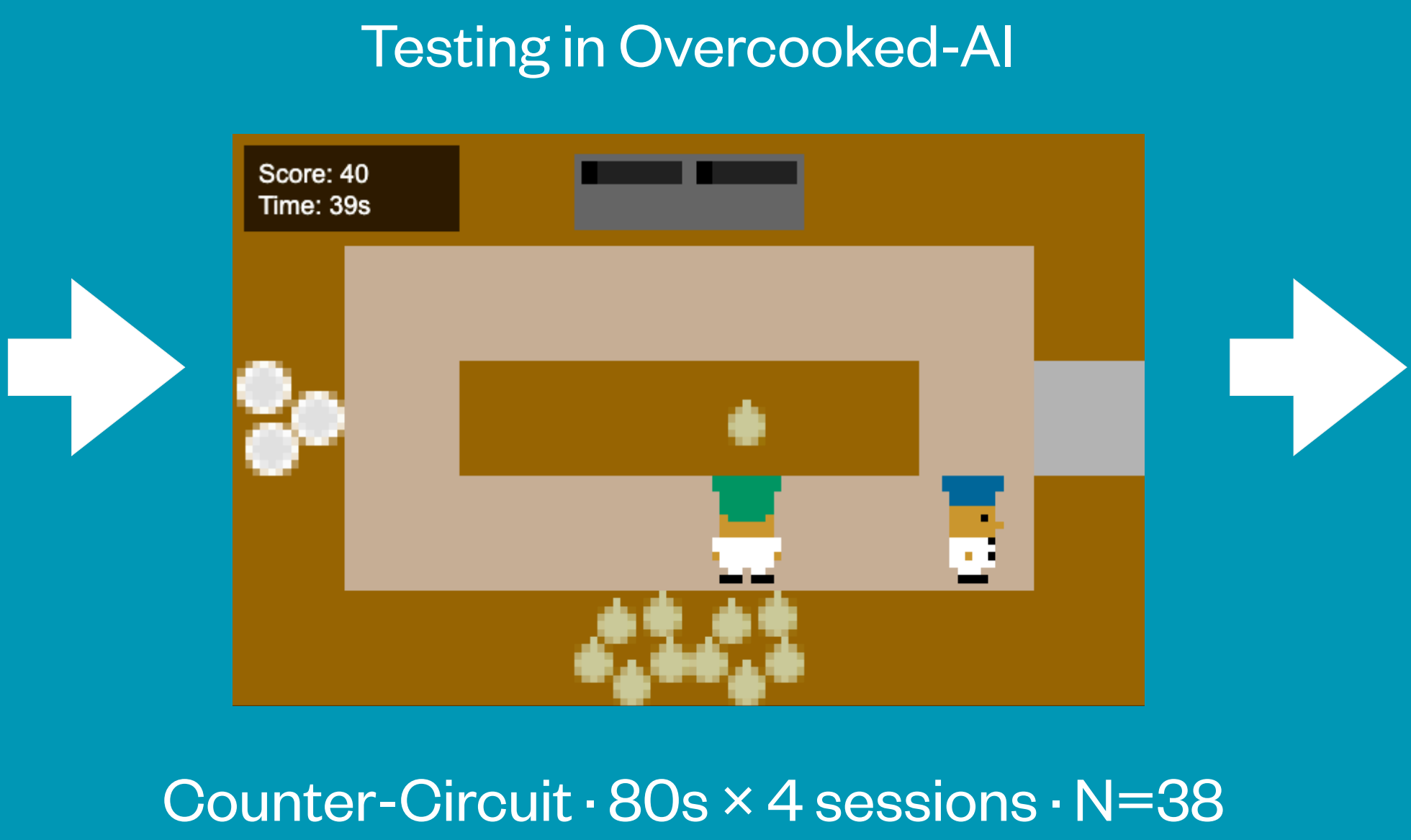
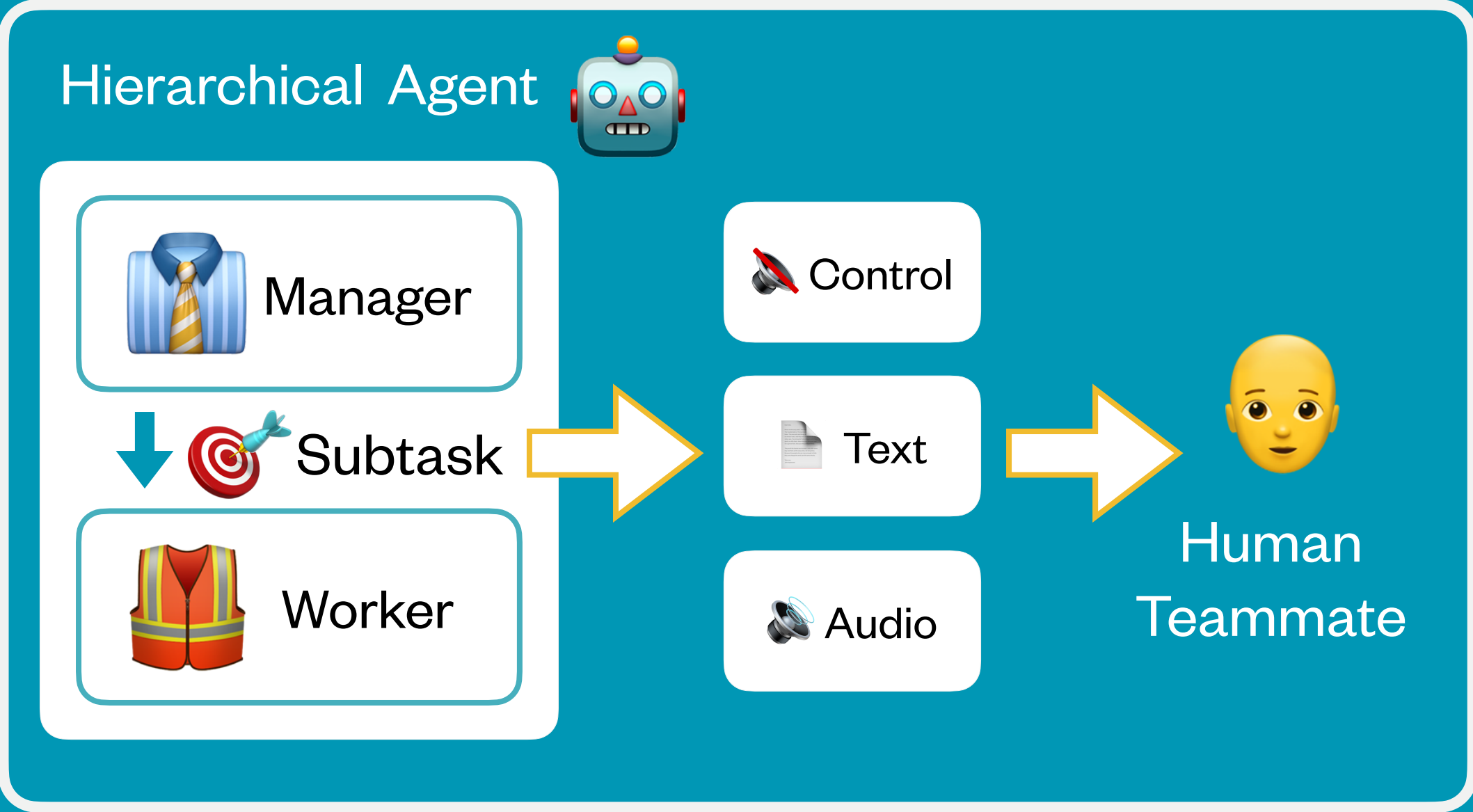


Evaluating XAI Support From A Hierarchical Reinforcement Learning Policy in Human-Agent Collaboration

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Equivalent performance

Explanations didn't raise peak score

2× faster adaptation

Audio participants adapted faster than Control

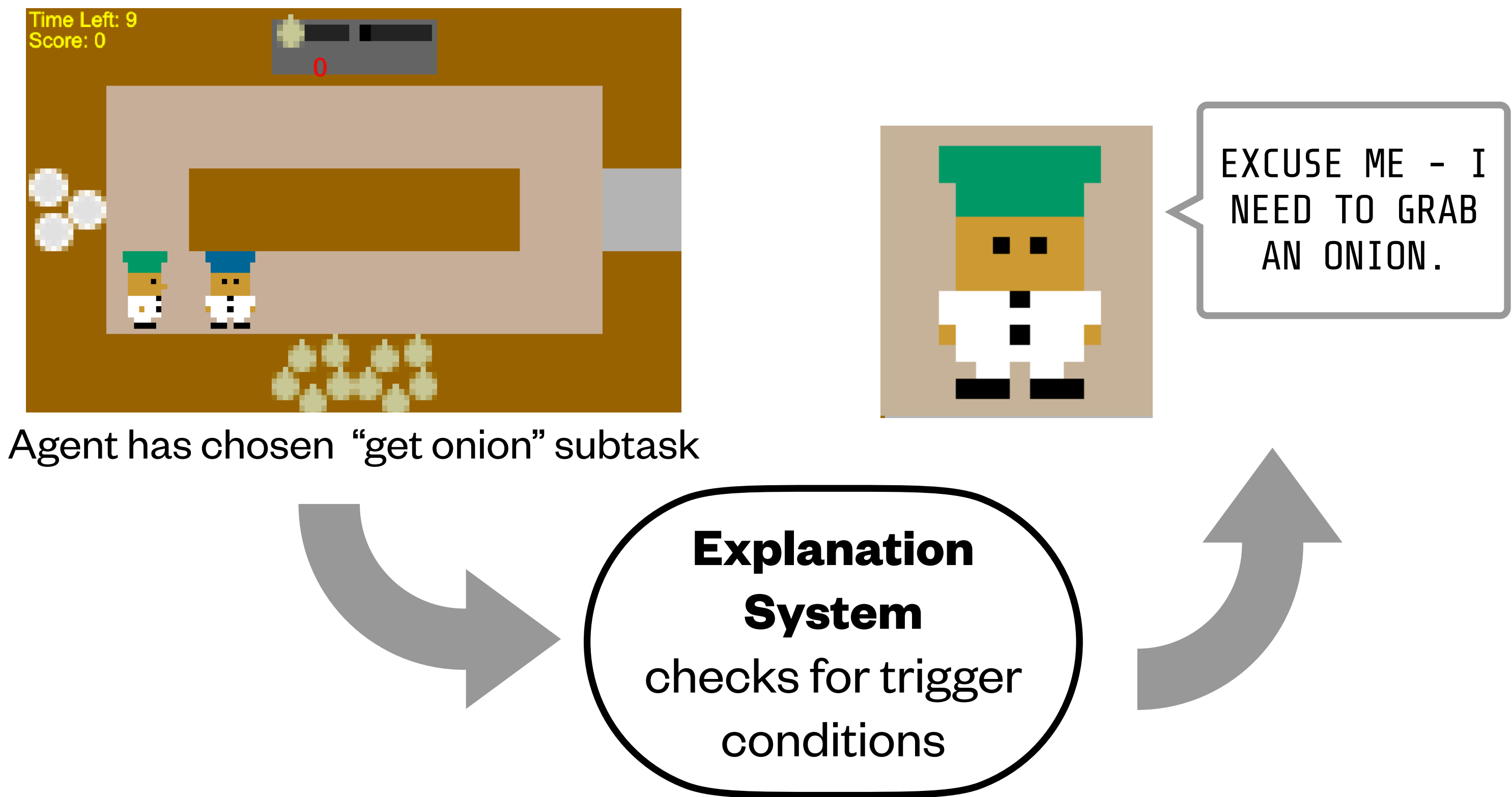
Weaker bond

Audio participants bonded less with the agent

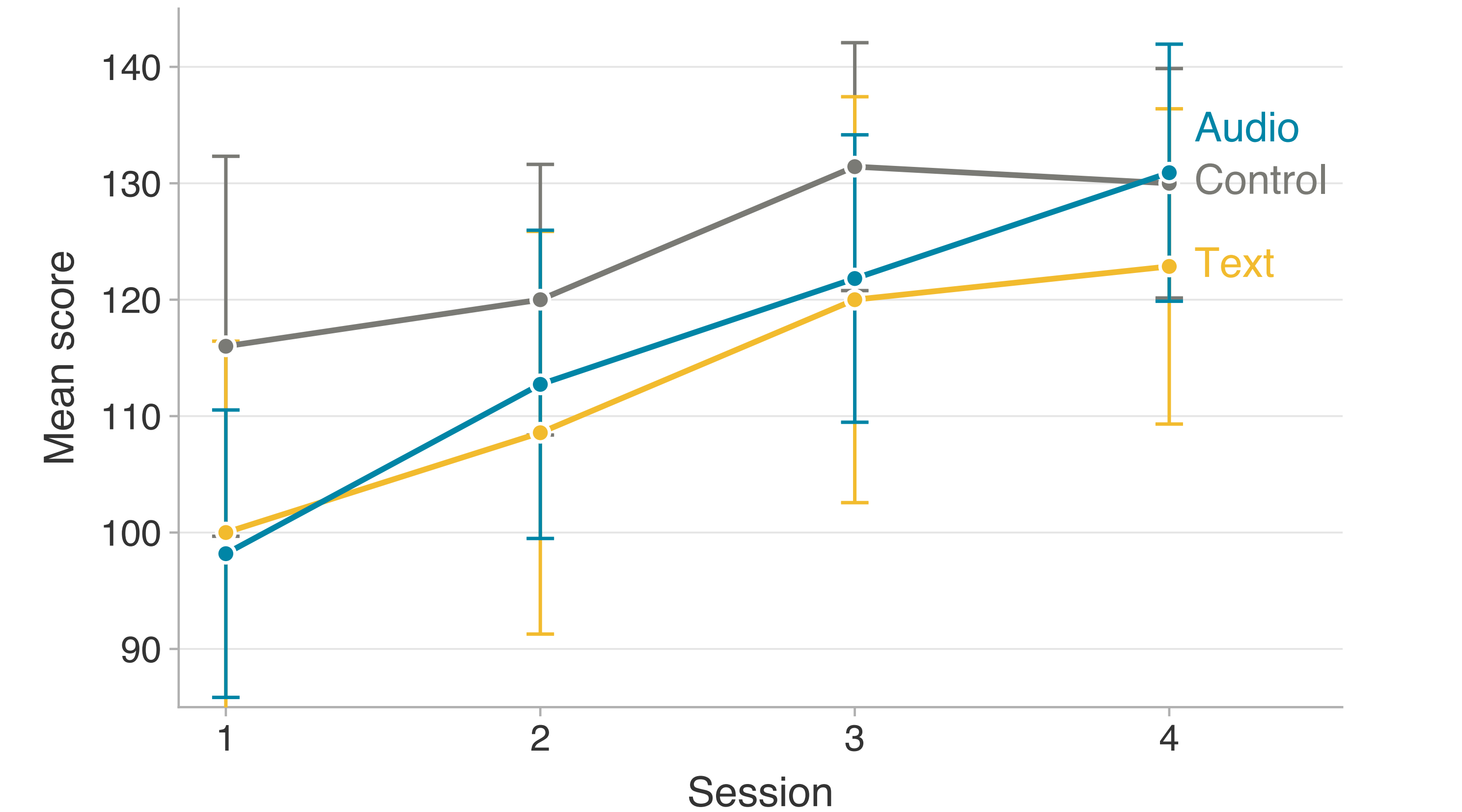
Real-time explanations didn't improve gaming experts' performance when teaming with agents — but audio explanations made them adapt 2× faster.

- First systematic XAI evaluation from a *learned* (through hierarchical RL) policy in a benchmark HAT environment
- Qualitative evidence showed audio explanations were more salient than text without higher cognitive load
- Higher transparency exposed the agent's artificial nature, with audio participants reporting weaker working alliances

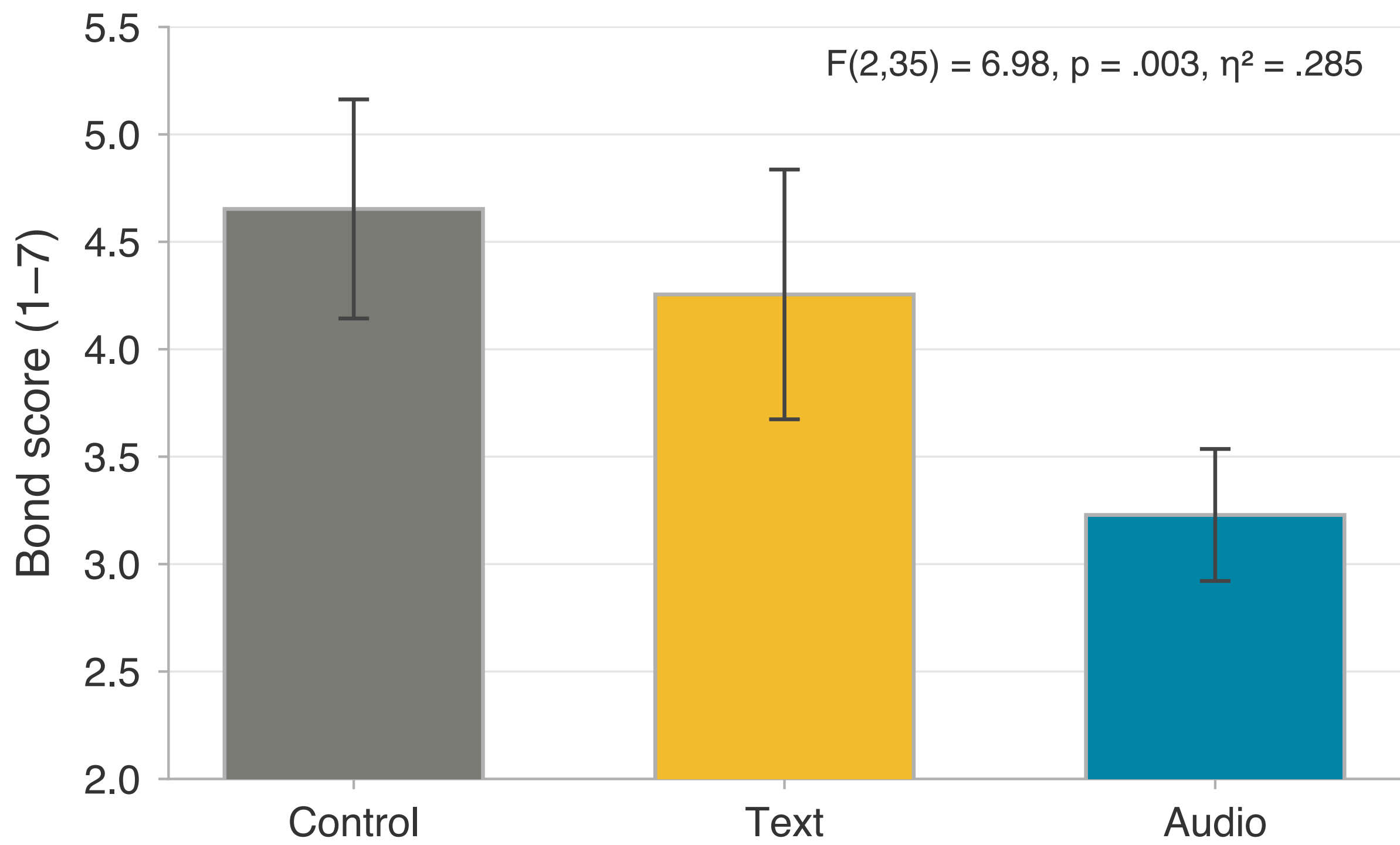
1 A trigger-based system delivers explanations from the agent's hierarchical decisions based on coordination opportunities.



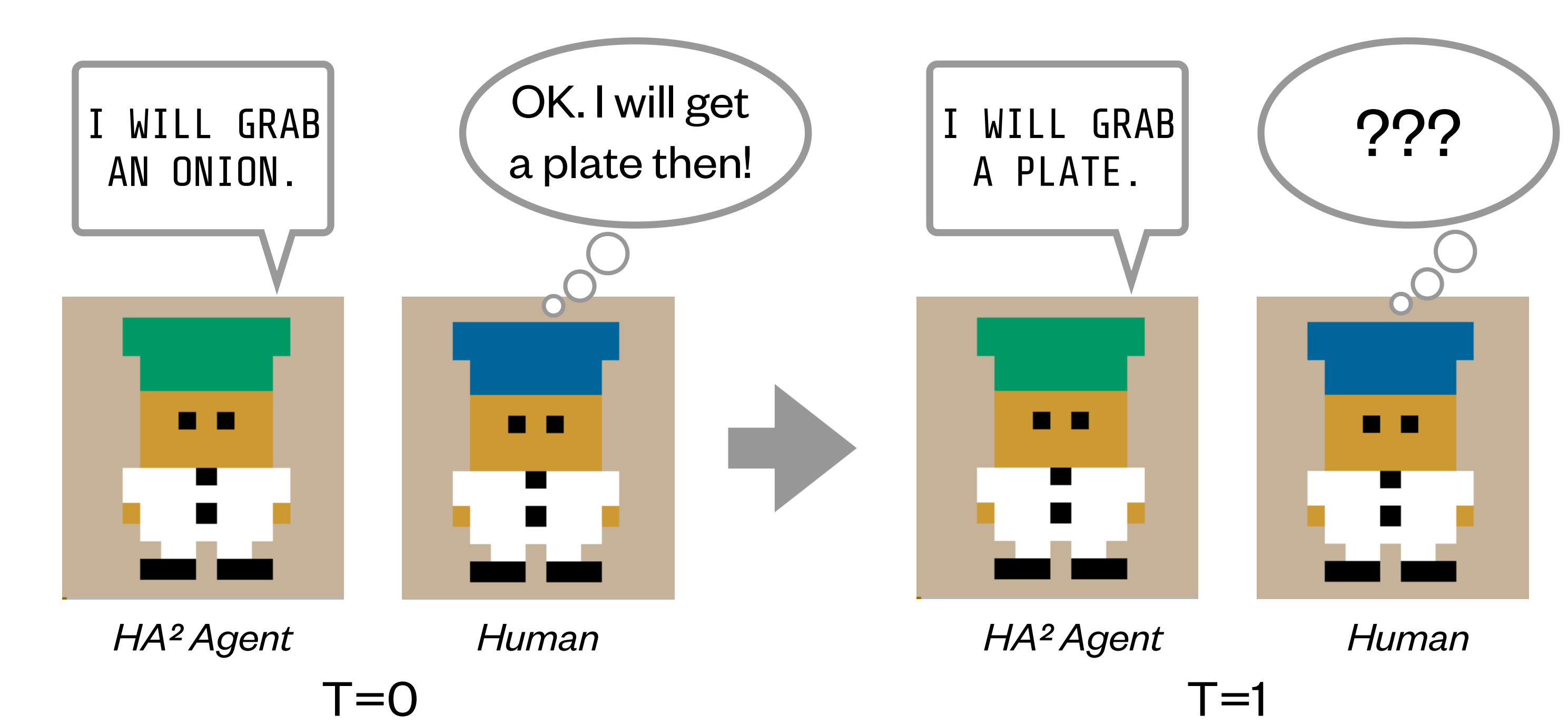
2 While explanations didn't raise peak performance in our high-experience sample, Audio participants adapted 2× faster than Control.



3 Even so, audio explanations weakened bonding with the agent, with anthropomorphism trending in the same direction.



4 The Manager switches subtasks mid-execution; audio cues are harder to ignore than text — likely driving the bonding cost.



Abstract

Explainable AI (XAI) research in human-agent collaboration has relied on hand-crafted policies in custom environments, limiting generalizability to state-of-the-art teaming benchmarks. We provide the first systematic evaluation of XAI support generated from a learned, intrinsically explainable policy in an established benchmark environment. We generate real-time explanations from hierarchical subtask selections via a trigger-based delivery system, comparing text versus audio modalities in a between-subjects experiment with gaming-experienced participants. While we found no statistically significant performance benefits, preliminary patterns suggest expertise may moderate explanation utility. This work establishes a methodological foundation for evaluating intrinsically explainable reinforcement learning in collaborative contexts, along with the first comparison of explanation modality effects in real-time human-agent teaming.



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